

1. Let the position vectors of the points A, B, C and D be $5\hat{i} + 5\hat{j} + 2\lambda\hat{k}$, $\hat{i} + 2\hat{j} + 3\hat{k}$, $-2\hat{i} + \lambda\hat{j} + 4\hat{k}$ and $-\hat{i} + 5\hat{j} + 6\hat{k}$. Let the set $S = \{\lambda \in \mathbb{R} : \text{the points } A, B, C \text{ and } D \text{ are coplanar}\}$. The $\sum_{\lambda \in S} (\lambda + 2)^2$ is equal to

[2023 (06 Apr Shift 1)]

- (1) 25
(2) $\frac{37}{2}$
(3) 14
(4) 41

2. Let $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} - 2\hat{j} - 2\hat{k}$ and $\vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{b}$ and $\vec{c}$, and $\vec{a} \cdot \vec{d} = 18$, then $|\vec{a} \times \vec{d}|^2$ is equal to

[2023 (06 Apr Shift 1)]

- (1) 640
(2) 680
(3) 720
(4) 760

3. Let the vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ represent three coterminal edges of a parallelepiped of volume V . Then the volume of the parallelepiped, whose coterminal edges are represented by $\vec{a}$, $\vec{b} + \vec{c}$ and $\vec{a} + 2\vec{b} + 3\vec{c}$ is equal to

[2023 (06 Apr Shift 2)]

- (1) $2V$
(2) $6V$
(3) V
(4) $3V$

4. The sum of all values of α , for which the points whose position vectors are $\hat{i} - 2\hat{j} + 3\hat{k}$, $2\hat{i} - 3\hat{j} + 4\hat{k}$, $(\alpha + 1)\hat{i} + 2\hat{k}$ and $9\hat{i} + (\alpha - 8)\hat{j} + 6\hat{k}$ are coplanar, is equal to

[2023 (06 Apr Shift 2)]

- (1) -2
(2) 2
(3) 6
(4) 4

5. If the points with position vectors $\alpha\hat{i} + 10\hat{j} + 13\hat{k}$, $6\hat{i} + 11\hat{j} + 11\hat{k}$, $\frac{9}{2}\hat{i} + \beta\hat{j} - 8\hat{k}$ are collinear, then $(19\alpha - 6\beta)^2$ is equal to

[2023 (08 Apr Shift 1)]

- (1) 36
(2) 25
(3) 49
(4) 16

6. Let $\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}$, $\vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k}$ and $\vec{c}$ be vectors such that $\vec{a} \times \vec{c} = \vec{a} \times \vec{b}$. If $\vec{a} \cdot \vec{c} = -12$, and $\vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$ then $\vec{c} \cdot (\hat{i} + \hat{j} + \hat{k})$ is equal to _____

[2023 (08 Apr Shift 1)]

7. Let the vectors $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$, and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ be coplanar. If the vectors $\vec{v}_1 = (a + b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b + c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c + a)\hat{k}$ are also coplanar, then $6(a + b + c)$ is equal to

[2023 (08 Apr Shift 2)]

- (1) 0
(2) 4
(3) 12
(4) 6

8. An arc PQ of a circle subtends a right angle at its centre O . The mid point of the arc PQ is R . If $\vec{OP} = \vec{u}$, $\vec{OR} = \vec{v}$ and $\vec{OQ} = \alpha\vec{u} + \beta\vec{v}$, then α, β^2 , are the roots of the equation

[2023 (10 Apr Shift 1)]

- (1) $x^2 + x - 2 = 0$
(2) $x^2 - x - 2 = 0$
(3) $3x^2 - 2x - 1 = 0$
(4) $3x^2 + 2x - 1 = 0$

9. Let $\vec{a} = 2\hat{i} + 7\hat{j} - \hat{k}$, $\vec{b} = 3\hat{i} + 5\hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$, and $\vec{c} \cdot \vec{d} = 12$. Then $(-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d})$ is equal to

[2023 (10 Apr Shift 2)]

- (1) 24
- (2) 44
- (3) 42
- (4) 48

10. For any vector $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$, with $10|a_i| < 1$, $i = 1, 2, 3$, consider the following statements:

(A) : $\max\{|a_1|, |a_2|, |a_3|\} \leq |\vec{a}|$

(B) : $|\vec{a}| \leq 3 \max\{|a_1|, |a_2|, |a_3|\}$

[2023 (11 Apr Shift 1)]

- (1) Only (B) is true
- (2) Only (A) is true
- (3) Both (A) and (B) are true
- (4) Neither (A) nor (B) is true

11. Let $\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}$, $\hat{i} + \hat{k}$ and $\hat{i} - \hat{j}$, $\hat{j} - \hat{k}$. If θ is the angle between the vector $\vec{a}$ and the vector $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{a} \cdot \vec{b} = 6$, then the ordered pair $(\theta, |\vec{a} \times \vec{b}|)$ is equal to

[2023 (11 Apr Shift 1)]

- (1) $(\frac{\pi}{3}, 3\sqrt{6})$
- (2) $(\frac{\pi}{4}, 3\sqrt{6})$
- (3) $(\frac{\pi}{3}, 6)$
- (4) $(\frac{\pi}{4}, 6)$

12. If four distinct points with position vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$ are coplanar, then $\left[\vec{a} \vec{b} \vec{c} \right]$ is equal to

[2023 (11 Apr Shift 2)]

- (1) $\left[\vec{a} \vec{b} \vec{a} \right] + \left[\vec{a} \vec{c} \vec{d} \right] + \left[\vec{d} \vec{b} \vec{c} \right]$
- (2) $\left[\vec{a} \vec{d} \vec{b} \right] + \left[\vec{d} \vec{c} \vec{a} \right] + \left[\vec{d} \vec{b} \vec{c} \right]$
- (3) $\left[\vec{d} \vec{c} \vec{a} \right] + \left[\vec{b} \vec{d} \vec{a} \right] + \left[\vec{c} \vec{d} \vec{b} \right]$
- (4) $\left[\vec{b} \vec{c} \vec{d} \right] + \left[\vec{d} \vec{a} \vec{c} \right] + \left[\vec{d} \vec{b} \vec{a} \right]$

13. Let $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{b} = \hat{i} + \hat{j} - \hat{k}$. If $\vec{c}$ is a vector such that $\vec{a} \cdot \vec{c} = 11$, $\vec{b} \cdot (\vec{a} \times \vec{c}) = 27$ and $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$, then $|\vec{a} \times \vec{c}|^2$ is equal to

[2023 (11 Apr Shift 2)]

14. Let a , b , c be three distinct real numbers, none equal to one. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + c\hat{k}$ are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is equal to

[2023 (12 Apr Shift 1)]

- (1) 2
- (2) -1
- (3) -2
- (4) 1

15. Let $\lambda \in \mathbb{Z}$, $\vec{a} = \lambda\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$. Let $\vec{c}$ be a vector such that $(\vec{a} + \vec{b} + \vec{c}) \times \vec{c} = \vec{0}$, $\vec{a} \cdot \vec{c} = -17$ and $\vec{b} \cdot \vec{c} = -20$. Then

$|\vec{c} \times (\lambda\hat{i} + \hat{j} + \hat{k})|^2$ is equal to

[2023 (12 Apr Shift 1)]

- (1) 46
- (2) 53
- (3) 62
- (4) 49

16. Let $\vec{a} = \hat{i} + 4\hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} + 7\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. If a vector $\vec{d}$ satisfies $\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$ and $\vec{d} \cdot \vec{a} = 24$, then $|\vec{d}|^2$ is equal to

[2023 (13 Apr Shift 1)]

- (1) 323
(2) 423
(3) 313
(4) 413

17. Let $\vec{a} = 3\hat{i} + \hat{j} - \hat{k}$ and $\vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}$. If $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $|\vec{b}|^2 = 50$, then $|\vec{b} + \vec{c}|^2$ is equal to _____.

[2023 (13 Apr Shift 1)]

18. Let $|\vec{a}| = 2$, $|\vec{b}| = 3$ and the angle between the vectors $\vec{a}$ and $\vec{b}$ be $\frac{\pi}{4}$. Then $\left| \left(\vec{a} + 2\vec{b} \right) \times \left(2\vec{a} - 3\vec{b} \right) \right|^2$ is equal to

[2023 (13 Apr Shift 2)]

- (1) 441
(2) 482
(3) 841
(4) 882

19. Let S be the set of all (λ, μ) for which the vectors $\lambda\hat{i} - \hat{j} + \hat{k}$, $\hat{j} + 2\hat{j} + \mu\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$, where $\lambda - \mu = 5$, are coplanar, then $\sum_{(\lambda, \mu) \in S} 80(\lambda^2 + \mu^2)$ is equal to

[2023 (15 Apr Shift 1)]

- (1) 2210
(2) 2130
(3) 2290
(4) 2370

20. Let $ABCD$ be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\vec{AB} - \vec{BC}) + (\vec{AD} - \vec{DC}) = k \vec{FE}$, then k is equal to

[2023 (15 Apr Shift 1)]

- (1) 4
(2) -2
(3) 2
(4) -4

ANSWER KEYS

1. (4) 2. (3) 3. (3) 4. (2) 5. (1) 6. (11) 7. (3) 8. (2)
9. (2) 10. (3) 11. (4) 12. (3) 13. (285) 14. (4) 15. (1) 16. (4)

1. (4)

Given that:

$A(5, 5, 2\lambda)$, $B(1, 2, 3)$, $C(-2, \lambda, 4)$ and $D(-1, 5, 6)$.

Now,

$$\vec{AB} \equiv (-4, -3, 3 - 2\lambda)$$

$$\vec{AC} \equiv (-7, \lambda - 5, 4 - 2\lambda)$$

$$\vec{AD} \equiv (-6, 0, 6 - 2\lambda)$$

Given that A, B, C and D are coplanar.

$$\therefore [\vec{AB} \vec{AC} \vec{AD}] = 0$$

$$\Rightarrow \begin{vmatrix} -4 & -3 & 3 - 2\lambda \\ -7 & \lambda - 5 & 4 - 2\lambda \\ -6 & 0 & 6 - 2\lambda \end{vmatrix} = 0$$

Expand the determinant along the first row.

$$\Rightarrow -4[(\lambda - 5)(6 - 2\lambda) - 0] - (-3)[-7(6 - 2\lambda) - (-6)(4 - 2\lambda)] + (3 - 2\lambda)[0 - (-6)(\lambda - 5)]$$

On solving the above equation we get,

$$\therefore \lambda = 3, \lambda = 2$$

We have two values of λ .

$$\Rightarrow \sum_{\lambda \in S} (\lambda + 2)^2 = (3 + 2)^2 + (2 + 2)^2$$

$$= 41$$

$$\text{Hence, } \sum_{\lambda \in S} (\lambda + 2)^2 = 41$$

2. (3)

Given that,

$$\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}, \vec{b} = \hat{i} - 2\hat{j} - 2\hat{k} \text{ and } \vec{c} = -\hat{i} + 4\hat{j} + 3\hat{k}.$$

Let us find $\vec{b} \times \vec{c}$

$$\Rightarrow \vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & -2 \\ -1 & 4 & 3 \end{vmatrix}$$

$$= (-6 + 8)\hat{i} - (3 - 2)\hat{j} + (4 - 2)\hat{k}$$

$$\Rightarrow \vec{b} \times \vec{c} = 2\hat{i} - \hat{j} + 2\hat{k}$$

Since $\vec{d}$ is perpendicular to both $\vec{b}$ and $\vec{c}$

$$\therefore \vec{d} = \lambda(2\hat{i} - \hat{j} + 2\hat{k})$$

$$\text{Also } \vec{a} \cdot \vec{d} = 18$$

$$\Rightarrow \lambda(2\hat{i} + 3\hat{j} + 4\hat{k}) \cdot (2\hat{i} - \hat{j} + 2\hat{k}) = 18$$

$$\Rightarrow 9\lambda = 18$$

$$\Rightarrow \lambda = 2$$

Now let us apply LaGrange's identity which is as follows,

$$\therefore |\vec{a} \times \vec{d}|^2 = |\vec{a}|^2 \cdot |\vec{d}|^2 - (\vec{a} \cdot \vec{d})^2$$

$$= |2\hat{i} + 3\hat{j} + 4\hat{k}|^2 |4\hat{i} - 2\hat{j} + 4\hat{k}|^2 - (18)^2$$

$$= 720$$

3. (3)

We know that,

Volume of parallelepiped whose edges determined by vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ is $\left[\vec{a} \vec{b} \vec{c} \right]$.

Now it is given that V is volume of parallelepiped $\vec{a}$, $\vec{b}$, $\vec{c}$, so required volume of parallelepiped with new edges will be

$$\begin{aligned} &= \left[\vec{a} \vec{b} + \vec{c} \vec{a} + 2\vec{b} + 3\vec{c} \right] \\ &= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 3 \end{vmatrix} \left[\vec{a} \vec{b} \vec{c} \right] \\ &= \{(1)(3-2)-0+0\} \left[\vec{a} \vec{b} \vec{c} \right] \\ &= \left[\vec{a} \vec{b} \vec{c} \right] \\ &= V \end{aligned}$$

Hence this is the correct option.

4. (2)

Let the given vectors be $\vec{A} = \hat{i} - 2\hat{j} + 3\hat{k}$, $\vec{B} = 2\hat{i} - 3\hat{j} + 4\hat{k}$, $\vec{C} = (\alpha + 1)\hat{i} + 2\hat{k}$ and $\vec{D} = 9\hat{i} + (\alpha - 8)\hat{j} + 6\hat{k}$

We know that if the vectors are coplanar then $\left[\vec{AB} \vec{AC} \vec{AD} \right] = 0$

$$\begin{aligned} \Rightarrow \vec{AB} &= (2\hat{i} - 3\hat{j} + 4\hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k}) = \hat{i} - \hat{j} + \hat{k} \\ \Rightarrow \vec{AC} &= ((\alpha + 1)\hat{i} + 2\hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k}) = \alpha\hat{i} + 2\hat{j} - \hat{k} \\ \Rightarrow \vec{AD} &= (9\hat{i} + (\alpha - 8)\hat{j} + 6\hat{k}) - (\hat{i} - 2\hat{j} + 3\hat{k}) = 8\hat{i} + (\alpha - 6)\hat{j} + 3\hat{k} \end{aligned}$$

Now,

$$\Rightarrow \begin{vmatrix} 1 & -1 & 1 \\ \alpha & 2 & -1 \\ 8 & \alpha - 6 & 3 \end{vmatrix} = 0$$

$$\Rightarrow 1(6 + \alpha - 6) + (3\alpha + 8) + (\alpha^2 - 6\alpha - 16) = 0$$

On Simplifying we get,

$$\Rightarrow \alpha^2 - 2\alpha - 8 = 0$$

Therefore, sum of the roots is $-(-2) = 2$.

5. (1)

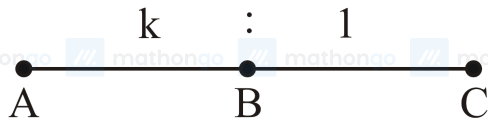
The given position vectors can be written as

- $A(\alpha, 10, 13)$
- $B(6, 11, 11)$
- $C\left(\frac{9}{2}, \beta, -8\right)$

Also given that these three points are collinear.

Let us assume that the point B divides AC in the ratio $k : 1$.

Since, A, B, C are collinear



On applying section formula for z co-ordinate we get,

$$11 = \frac{-8k + 13}{k + 1}$$

$$\Rightarrow 11k + 11 = -8k + 13$$

$$\Rightarrow 19k = 2$$

$$\Rightarrow k = \frac{2}{19}$$

$$\therefore \text{Ratio} = 2 : 19$$

$$\text{Now, } \frac{\alpha \times 19 + \left(\frac{9}{2}\right) \times 2}{2 + 19} = 6$$

$$\Rightarrow 19\alpha = 117$$

$$\Rightarrow \alpha = \frac{117}{19}$$

$$\text{Now similarly, } \frac{2\beta + 190}{21} = 11$$

$$\Rightarrow \beta = \frac{41}{2}$$

$$\therefore (19\alpha - 6\beta)^2 = (117 - 123)^2 = 36$$

Hence this is the correct option.

6. (11)

Given,

$$\vec{a} = 6\hat{i} + 9\hat{j} + 12\hat{k}, \vec{b} = \alpha\hat{i} + 11\hat{j} - 2\hat{k} \text{ and } \vec{c} \text{ be vectors such that } \vec{a} \times \vec{c} = \vec{a} \times \vec{b},$$

$$\vec{a} \times \vec{c} - \vec{a} \times \vec{b} = 0$$

$$\Rightarrow \vec{a} \times (\vec{c} - \vec{b}) = 0$$

So, $\vec{a}$ & $(\vec{c} - \vec{b})$ are parallel vectors,

$$\text{Hence, } \lambda \vec{a} = \vec{c} - \vec{b}$$

$$\Rightarrow \vec{c} = \vec{b} + \lambda \vec{a}$$

$$\Rightarrow \vec{a} \cdot \vec{c} = \vec{a} \cdot \vec{b} + \lambda |\vec{a}|^2$$

$$\Rightarrow -12 = (6\alpha + 75) + \lambda(261)$$

$$\Rightarrow 2\alpha + 87\lambda = -29 \dots (i)$$

Now again using $\vec{c} = \vec{b} + \lambda \vec{a}$ we get,

$$\vec{c} = \hat{i}(\alpha + 6\lambda) + \hat{j}(11 + 9\lambda) + \hat{k}(-2 + 12\lambda)$$

$$\text{Also given } \vec{c} \cdot (\hat{i} - 2\hat{j} + \hat{k}) = 5$$

$$\Rightarrow (\alpha + 6\lambda) - 2(11 + 9\lambda) + (-2 + 12\lambda) = 5$$

$$\Rightarrow \alpha = 29$$

$$\text{So, } 2\alpha + 87\lambda = -29$$

$$\Rightarrow \lambda = -1$$

$$\text{Hence, } \vec{c} = 23\hat{i} + 2\hat{j} - 14\hat{k}$$

$$\text{So, the value of } \vec{c} \cdot (\hat{i} + \hat{j} + \hat{k}) = 23 + 2 - 14 = 11$$

7. (3)

Given: $\vec{u}_1 = \hat{i} + \hat{j} + a\hat{k}$, $\vec{u}_2 = \hat{i} + b\hat{j} + \hat{k}$ and $\vec{u}_3 = c\hat{i} + \hat{j} + \hat{k}$ are coplanar.

Also given that, $\vec{v}_1 = (a+b)\hat{i} + c\hat{j} + c\hat{k}$, $\vec{v}_2 = a\hat{i} + (b+c)\hat{j} + a\hat{k}$ and $\vec{v}_3 = b\hat{i} + b\hat{j} + (c+a)\hat{k}$ are coplanar.

Now, using the condition of coplanar we get,

$$\Rightarrow \begin{vmatrix} 1 & 1 & a \\ 1 & b & 1 \\ c & 1 & 1 \end{vmatrix} = 0$$

Expanding the determinant along R_1 .

$$\Rightarrow (b-1) - (1-c) + a(1-bc) = 0$$

$$\Rightarrow a + b + c = 2 + abc \quad \dots (i)$$

Again using the coplanar condition we get,

$$\Rightarrow \begin{vmatrix} a+b & c & c \\ a & b+c & a \\ b & b & c+a \end{vmatrix} = 0$$

Apply row transformations, ($R_3 \rightarrow R_3 - (R_1 + R_2)$)

$$\Rightarrow \begin{vmatrix} a+b & c & c \\ a & b+c & a \\ -2a & -2c & 0 \end{vmatrix} = 0$$

Expand the determinant along R_1 .

$$\Rightarrow (a+b)(0+2ac) - c(0+2a^2) + c(-2ac+2a(b+c)) = 0$$

$$\Rightarrow 2a^2c + 2abc - 2a^2c - 2ac^2 + 2abc + 2ac^2 = 0$$

$$\Rightarrow abc = 0$$

$$\therefore a + b + c = 2 \text{ (From eq (i))}$$

$$\therefore 6(a+b+c) = 12$$

Hence this is the correct option.

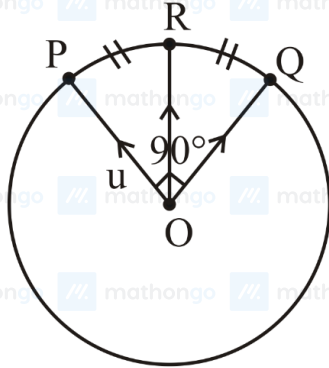
8. (2)

Given,

An arc PQ of a circle subtends a right angle at its centre O . The mid point of the arc PQ is R ,

And $\vec{OP} = \vec{u}$, $\vec{OR} = \vec{v}$ and $\vec{OQ} = \alpha \vec{u} + \beta \vec{v}$,

So, plotting the diagram of the given value we get,



Now given, $\vec{OQ} = \alpha \vec{u} + \beta \vec{v}$

Where, $\vec{OR} = \vec{v}$ and $\vec{OP} = \vec{u}$

Now $|\vec{OP}| = |\vec{OQ}| = \text{radius}$, so R will lie on angle bisector of $\vec{OQ}$ and $\vec{OP}$

Now using the perpendicular condition we get,

$$\vec{OQ} \cdot \vec{OP} = 0$$

$$\Rightarrow \vec{OQ} \cdot \vec{OP} = \alpha |\vec{u}|^2 + \beta (\vec{v} \cdot \vec{u}) = 0$$

$$\Rightarrow \alpha |\vec{u}|^2 + \beta (|\vec{v}| |\vec{u}| \cos 45^\circ) = 0$$

$$\Rightarrow \alpha + \beta \cdot \cos 45^\circ = 0 \left\{ \text{as } |\vec{u}| = |\vec{v}| = \text{radius} \right\}$$

$$\Rightarrow \alpha = \frac{-\beta}{\sqrt{2}}$$

$$\text{Now solving, } \vec{OQ} \cdot \vec{OR} = |\vec{OQ}| |\vec{OR}| \cos 45^\circ = \frac{1}{\sqrt{2}} r^2$$

$$\Rightarrow (\alpha \vec{v} + \beta \vec{v}) \cdot (\vec{v}) = \frac{r^2}{\sqrt{2}}$$

$$\Rightarrow \alpha \cdot \frac{r^2}{\sqrt{2}} + \beta \cdot r^2 = \frac{r^2}{\sqrt{2}}$$

$$\Rightarrow \frac{\alpha}{\sqrt{2}} + \beta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \beta = \sqrt{2} \left\{ \text{as } \alpha = \frac{-\beta}{\sqrt{2}} \right\}$$

So, $\alpha = -1$

Now finding the equation with roots $\alpha = -1$, $\beta^2 = 2$ we get, $x^2 - x - 2 = 0$

9. (2)

Given,

$$\vec{a} = 2\hat{i} + 7\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 5\hat{k} \text{ and } \vec{c} = \hat{i} - \hat{j} + 2\hat{k}$$

And $\vec{d}$ be a vector which is perpendicular to both $\vec{a}$ and $\vec{b}$,

$$\text{So, } \vec{d} = \lambda(\vec{a} \times \vec{b})$$

$$\text{Now finding, } \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 7 & -1 \\ 3 & 0 & 5 \end{vmatrix}$$

$$\Rightarrow \vec{a} \times \vec{b} = 35\hat{i} - 13\hat{j} - 21\hat{k}$$

$$\text{So, } \vec{d} = \lambda(35\hat{i} - 13\hat{j} - 21\hat{k})$$

Also given,

$$\vec{c} \cdot \vec{d} = 12$$

$$\Rightarrow \vec{c} \cdot \vec{d} = (\hat{i} - \hat{j} + 2\hat{k}) \cdot \lambda(35\hat{i} - 13\hat{j} - 21\hat{k})$$

$$\Rightarrow \lambda(35 + 13 - 42) = 12$$

$$\Rightarrow \lambda = 2$$

$$\text{So, } \vec{c} \times \vec{d} = 2 \times \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 2 \\ 35 & -13 & -21 \end{vmatrix}$$

$$\Rightarrow \vec{c} \times \vec{d} = 2(47\hat{i} + 91\hat{j} + 22\hat{k})$$

Now finding,

$$(-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d}) = 2(-\hat{i} + \hat{j} - \hat{k}) \cdot (47\hat{i} + 91\hat{j} + 22\hat{k})$$

$$\Rightarrow (-\hat{i} + \hat{j} - \hat{k}) \cdot (\vec{c} \times \vec{d}) = 2(-47 + 91 - 22) = 44$$

10. (3)

Given,

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}, \text{ with } 10|a_i| < 1, i = 1, 2, 3,$$

Let us assume that $|a_1| \leq |a_2| \leq |a_3|$

Now we know that,

$$|\vec{a}|^2 = |a_1|^2 + |a_2|^2 + |a_3|^2$$

$$\text{And } |\vec{a}_3|^2 \leq |a_1|^2 + |a_2|^2 + |a_3|^2$$

So, combining both above equations we get,

$$\Rightarrow |\vec{a}|^2 \geq |\vec{a}_3|^2$$

$$\Rightarrow |\vec{a}| \geq |\vec{a}_3| = \max\{|\vec{a}_1|, |\vec{a}_2|, |\vec{a}_3|\}$$

Hence, (A) is true

Now again solving,

$$|\vec{a}|^2 = |a_1|^2 + |a_2|^2 + |a_3|^2$$

$$\text{And } |a_1|^2 + |a_2|^2 + |a_3|^2 \leq |a_3|^2 + |a_3|^2 + |a_3|^2$$

$$\Rightarrow |a_1|^2 + |a_2|^2 + |a_3|^2 \leq 3|a_3|^2$$

$$\Rightarrow |\vec{a}|^2 \leq 3|a_3|^2$$

$$\Rightarrow |\vec{a}| \leq 3|a_3| = \max\{|\vec{a}_1|, |\vec{a}_2|, |\vec{a}_3|\}$$

Hence, (B) is also true.

11. (4)

Given,

$\vec{a}$ be a non-zero vector parallel to the line of intersection of the two planes described by $\hat{i} + \hat{j}, \hat{i} + \hat{k}$,

So finding the normal vector to the above plane, we get,

$$\vec{n}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = \hat{i} - \hat{j} - \hat{k}$$

And normal vector to planes $\hat{i} - \hat{j}, \hat{j} - \hat{k}$ will be,

$$\vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{vmatrix} = \hat{i} + \hat{j} + \hat{k}$$

$$\text{Now } \vec{a} = \lambda(\vec{n}_1 \times \vec{n}_2)$$

$$\text{So, direction ratio of } \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & -1 \\ 1 & 1 & 1 \end{vmatrix} = -2\hat{j} + 2\hat{k}$$

$$\text{Hence, direction ratio of } \vec{a} = (0, -2, 2) = (0, -1, 1)$$

$$\text{Now given, } \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\text{And } \vec{a} = \lambda(-\hat{j} + \hat{k})$$

$$\text{So, } \vec{a} \cdot \vec{b} = 6 = \lambda(2 + 1)$$

$$\Rightarrow \lambda = 2$$

$$\therefore \vec{a} = -2\hat{j} + 2\hat{k}$$

$$\text{Now we know that, } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$$

$$\Rightarrow 6 = 2\sqrt{2} \times 3 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

$$\text{Now finding, } |\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta = 2\sqrt{2} \times 3 \times \frac{1}{\sqrt{2}} = 6$$

$$\text{Hence, ordered pair } \left(\theta, |\vec{a} \times \vec{b}| \right) = \left(\frac{\pi}{4}, 6 \right)$$

12. (3)

Given that $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ are coplanar.

$\Rightarrow \vec{b} - \vec{a}, \vec{c} - \vec{a}, \vec{d} - \vec{a}$ are also coplanar.

$$\Rightarrow [\vec{b} - \vec{a}, \vec{c} - \vec{a}, \vec{d} - \vec{a}] = 0$$

$$\Rightarrow (\vec{b} - \vec{a}) \cdot ((\vec{c} - \vec{a}) \times (\vec{d} - \vec{a})) = 0$$

$$\Rightarrow (\vec{b} - \vec{a}) \cdot (\vec{c} \times \vec{d} - \vec{c} \times \vec{a} - \vec{a} \times \vec{d}) = 0 \quad (\text{Since } \vec{a} \times \vec{a} = 0)$$

$$\Rightarrow \vec{b} \cdot (\vec{c} \times \vec{d}) - \vec{b} \cdot (\vec{c} \times \vec{a}) - \vec{b} \cdot (\vec{a} \times \vec{d}) - \vec{a} \cdot (\vec{c} \times \vec{d}) + \vec{a} \cdot (\vec{c} \times \vec{a}) + \vec{a} \cdot (\vec{a} \times \vec{d}) = 0$$

$$\Rightarrow [\vec{b}, \vec{c}, \vec{d}] - [\vec{b}, \vec{c}, \vec{a}] - [\vec{b}, \vec{a}, \vec{d}] - [\vec{a}, \vec{c}, \vec{d}] = 0$$

$$\text{We know that } \begin{bmatrix} \vec{x} & \vec{y} & \vec{z} \end{bmatrix} = \begin{bmatrix} \vec{y} & \vec{z} & \vec{x} \end{bmatrix} = \begin{bmatrix} \vec{z} & \vec{x} & \vec{y} \end{bmatrix} \text{ and } \begin{bmatrix} \vec{x} & \vec{y} & \vec{z} \end{bmatrix} = -\begin{bmatrix} \vec{x} & \vec{z} & \vec{y} \end{bmatrix}$$

$$\Rightarrow [\vec{b}, \vec{c}, \vec{d}] + [\vec{a}, \vec{b}, \vec{d}] + [\vec{a}, \vec{d}, \vec{c}] = [\vec{a}, \vec{b}, \vec{c}]$$

$$\Rightarrow [\vec{c}, \vec{d}, \vec{b}] + [\vec{b}, \vec{d}, \vec{a}] + [\vec{d}, \vec{c}, \vec{a}] = [\vec{a}, \vec{b}, \vec{c}]$$

$$\text{Hence, the required answer is } \begin{bmatrix} \vec{d} & \vec{c} & \vec{a} \end{bmatrix} + \begin{bmatrix} \vec{b} & \vec{d} & \vec{a} \end{bmatrix} + \begin{bmatrix} \vec{c} & \vec{d} & \vec{b} \end{bmatrix}$$

13. (285)

Given,

$$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = \hat{i} + \hat{j} - \hat{k}, \vec{a} \cdot \vec{c} = 11, \vec{b} \cdot (\vec{a} \times \vec{c}) = 27 \text{ and } \vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$$

Now finding, $(\vec{b} \times \vec{a})$ we get,

$$(\vec{b} \times \vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{vmatrix} = 5\hat{i} - 4\hat{j} + \hat{k}$$

$$\text{Let } \vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$$

Now solving, $\vec{a} \cdot \vec{c} = 11$ we get,

$$c_1 + 2c_2 + 3c_3 = 11 \dots\dots\dots(1)$$

Now solving, $\vec{b} \cdot \vec{c} = -\sqrt{3}|\vec{b}|$ we get,

$$c_1 + c_2 - c_3 = -\sqrt{3}\sqrt{3}$$

$$c_1 + c_2 - c_3 = -3 \dots\dots\dots(2)$$

Now solving $(\vec{b} \times \vec{a}) \cdot \vec{c} = 27$ we get,

$$5c_1 - 4c_2 + c_3 = 27 \dots\dots\dots(3)$$

Now on solving equation (1), (2) & (3) we get,

$$\vec{c} = 3\hat{i} - 2\hat{j} + 4\hat{k}$$

$$\text{Hence, } |\vec{a} \times \vec{c}|^2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 3 & -2 & 4 \end{vmatrix}^2 = |14\hat{i} + 5\hat{j} - 8\hat{k}|^2$$

$$\Rightarrow |\vec{a} \times \vec{c}|^2 = 14^2 + 5^2 + 8^2 = 285$$

14. (4)

Given that the vectors are coplanar.

That means the determinant is equal to zero.

$$\Rightarrow \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$$\Rightarrow R_1 \rightarrow R_1 - R_2 \text{ and } R_2 \rightarrow R_2 - R_3$$

$$\Rightarrow \begin{vmatrix} a-1 & 1-b & 0 \\ 0 & b-1 & 1-c \\ 1 & 1 & c \end{vmatrix} = 0$$

Expand the determinant along C_1 .

$$\Rightarrow (a-1)[c(b-1) - (1-c)] + 1[(1-b)(1-c)] = 0$$

$$\Rightarrow c(a-1)(b-1) - (a-1)(1-c) + (1-b)(1-c) = 0$$

$$\Rightarrow c(1-a)(1-b) + (1-a)(1-c) + (1-b)(1-c) + (1-a)(1-b)(1-c) = (1-a)(1-b)(1-c)$$

$$\Rightarrow (1-a)(1-b) + (1-a)(1-c) + (1-b)(1-c) = (1-a)(1-b)(1-c)$$

$$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

Hence this is the required option.

15. (1)

Given that

$$\vec{a} = \lambda \hat{i} + \hat{j} - \hat{k}$$

$$\vec{b} = 3\hat{i} - \hat{j} + 2\hat{k}$$

$$\text{Also } (\vec{a} + \vec{b} + \vec{c}) \times \vec{c} = \vec{0}$$

$$\Rightarrow k(\vec{a} + \vec{b}) = \vec{c}$$

$$\vec{a} \cdot \vec{c} = -17 \text{ and } \vec{b} \cdot \vec{c} = -20$$

$$\text{Now, Take } \vec{a} \cdot \vec{c} = -17$$

$$\Rightarrow k(\lambda \hat{i} + \hat{j} - \hat{k}) \cdot (\lambda \hat{i} + \hat{j} - \hat{k} + 3\hat{i} - \hat{j} + 2\hat{k}) = -17$$

$$\Rightarrow k(\lambda^2 + 3\lambda + 0 - 1) = -17$$

$$\Rightarrow k(\lambda^2 + 3\lambda - 1) = -17 \dots (1)$$

Similarly, on taking $\vec{b} \cdot \vec{c} = -20$, we get,

$$k(3\hat{i} - \hat{j} + 2\hat{k}) \cdot (\lambda \hat{i} + \hat{j} - \hat{k} + 3\hat{i} - \hat{j} + 2\hat{k}) = -20$$

$$\Rightarrow k(3\lambda + 9 + 2) = -20$$

$$\Rightarrow k(3\lambda + 11) = -20 \dots (2)$$

Now on solving equation (1) & (2) we get,

$$\Rightarrow 20\lambda^2 + 9\lambda - 207 = 0$$

$$\Rightarrow \lambda = 3, \frac{-9}{20}$$

$$\text{For } \lambda = 3, k = -1$$

$$\Rightarrow \vec{c} = -1(\vec{a} + \vec{b})$$

$$\Rightarrow -((\lambda + 3)\hat{i} + \hat{k}) = -6\hat{i} - \hat{k}$$

$$\Rightarrow \vec{c} \times (\lambda \hat{i} + \hat{j} + \hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -6 & 0 & -1 \\ 3 & 1 & 1 \end{vmatrix} = \hat{i} - 3\hat{j} + 6\hat{k}$$

$$\Rightarrow |\vec{c} \times (\lambda \hat{i} + \hat{j} + \hat{k})|^2 = 46$$

Hence this is the correct option.

16. (4)

We have been given that

$$\vec{d} \times \vec{b} = \vec{c} \times \vec{b}$$

$$\Rightarrow (\vec{d} - \vec{c}) \times \vec{b} = \vec{0}$$

$$\Rightarrow \vec{d} - \vec{c} = \lambda \vec{b} \quad (\because \vec{d} \neq \vec{c})$$

$$\Rightarrow \vec{d} = \vec{c} + \lambda \vec{b}$$

$$\text{It is given that } \vec{a} \cdot \vec{d} = 24$$

$$\Rightarrow \vec{a} \cdot \vec{c} + \lambda \vec{b} \cdot \vec{a} = 24$$

$$\Rightarrow 6 + \lambda(3 - 8 + 14) = 24$$

$$\Rightarrow 9\lambda = 18$$

$$\Rightarrow \lambda = 2$$

$$\therefore \vec{d} = \vec{c} + 2\vec{b}$$

$$\Rightarrow \vec{d} = 8\hat{i} - 5\hat{j} + 18\hat{k}$$

$$|\vec{d}|^2 = 64 + 25 + 324 = 413$$

Hence, this is the correct option.

17. (66)

Given,

$$\vec{a} = 3\hat{i} + \hat{j} - \hat{k} \text{ and } \vec{c} = 2\hat{i} - 3\hat{j} + 3\hat{k}.$$

And $\vec{b}$ is a vector such that $\vec{a} = \vec{b} \times \vec{c}$ and $|\vec{b}|^2 = 50$,

Now solving, $|\vec{a}| = |\vec{b} \times \vec{c}|$

$$\Rightarrow \sqrt{3^2 + 1^2 + 1^2} = |\vec{b}| \cdot \sqrt{2^2 + 3^2 + 3^2} \cdot \sin \theta$$

$$\Rightarrow \sqrt{11} = \sqrt{50} \cdot \sqrt{22} \cdot \sin \theta$$

$$\therefore \sin \theta = \frac{1}{10} \text{ or } \cos \theta = \frac{\sqrt{99}}{10}$$

Now solving, $72 - |\vec{b} + \vec{c}|^2$ we get,

$$72 - |\vec{b} + \vec{c}|^2$$

$$= 72 - (|\vec{b}|^2 + |\vec{c}|^2 + 2|\vec{b}||\vec{c}|\cos \theta)$$

$$= 72 - (50 + 22 + 2 \times 5\sqrt{2} \cdot \sqrt{22} \cdot \frac{\sqrt{99}}{10})$$

$$= 72 - (72 + \frac{2 \times 5 \times 2 \times 11 \times 3}{10})$$

$$= |66| = 66$$

18. (4)

Given,

$$|\vec{a}| = 2, |\vec{b}| = 3 \text{ and the angle between the vectors } \vec{a} \text{ and } \vec{b} \text{ be } \frac{\pi}{4},$$

Now solving,

$$|(\vec{a} + 2\vec{b}) \times (2\vec{a} - 3\vec{b})|^2$$

$$= |-3(\vec{a} \times \vec{b}) + 4(\vec{b} \times \vec{a})|^2 \left\{ \text{as } \vec{a} \times \vec{a} = 0 \text{ \& } \vec{b} \times \vec{b} = 0 \right\}$$

$$= |7(\vec{b} \times \vec{a})|^2 \left\{ \text{as } \vec{b} \times \vec{a} = -\vec{a} \times \vec{b} \right\}$$

$$= 49|\vec{a}|^2|\vec{b}|^2 \sin^2 \frac{\pi}{4}$$

$$= 49 \times 4 \times 9 \times \frac{1}{2}$$

$$= 882$$

19. (3)

The given equations are coplanar.

$$\Rightarrow \begin{vmatrix} \lambda & -1 & 1 \\ 1 & 2 & \mu \\ 3 & -4 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \lambda[10 + 4\mu] + 1[5 - 3\mu] + 1[-10] = 0$$

$$\Rightarrow 4\lambda\mu + 10\lambda - 3\mu = 5 \dots\dots(i)$$

$$\text{And } \lambda - \mu = 5$$

So, by (1)

$$\Rightarrow 4(5 + \mu)\mu + 10(5 + \mu) - 3\mu = 5$$

$$\Rightarrow 4\mu^2 + 27\mu + 45 = 0$$

$$\Rightarrow 4\mu^2 + 15\mu + 12\mu + 45 = 0$$

$$\therefore \mu = -3, \frac{-15}{4}$$

$$\Rightarrow \lambda = 2, \frac{5}{4}$$

$$\therefore \sum_{\lambda, \mu \in S} 80(\lambda^2 + \mu^2) = 80 \left[(9 + 4) + \frac{225}{16} + \frac{25}{16} \right]$$

$$= 80 \left[13 + \frac{125}{8} \right] = 10 \times 229$$

$$= 2290$$

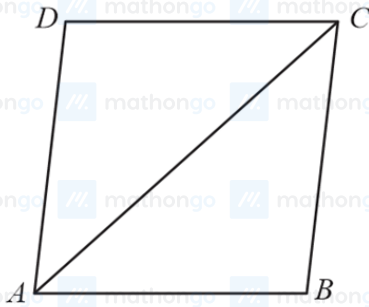
Hence this is the correct option.

20. (4)

Given,

$ABCD$ be a quadrilateral. If E and F are the mid points of the diagonals AC and BD respectively and $(\vec{AB} - \vec{BC}) + (\vec{AD} - \vec{DC}) = k \vec{FE}$,

Now plotting the diagram we get,



Now let position vector of A, B, C and D are $\vec{a}, \vec{b}, \vec{c}$ and $\vec{d}$ respectively

Now by midpoint formula,

$$\text{Position vector of } E = \frac{\vec{OC} + \vec{OA}}{2} = \frac{\vec{c} + \vec{a}}{2}$$

$$\text{And Position vector of } F = \frac{\vec{b} + \vec{d}}{2}$$

$$\text{Now, } (\vec{AB} - \vec{BC}) + \vec{AD} - \vec{DC}$$

$$\Rightarrow \vec{b} - \vec{a} - (\vec{c} - \vec{b}) + \vec{d} - \vec{a} - (\vec{c} - \vec{d})$$

$$\Rightarrow 2\vec{b} - 2\vec{a} - 2\vec{c} + 2\vec{d}$$

$$\Rightarrow 2(\vec{b} + \vec{d}) - 2(\vec{a} + \vec{c})$$

$$\Rightarrow 4\left[\frac{\vec{b} + \vec{d}}{2} - \frac{\vec{a} + \vec{c}}{2}\right] = 4[\vec{OF} - \vec{OE}]$$

$$\Rightarrow 4\vec{EF} = -4\vec{FE}$$

$$\therefore k = -4$$